**Motilal Nehru National Institute of Technology Allahabad**
Department of Electronics and Communication Engineering

DIGITAL SIGNAL PROCESSING · COURSE PROJECT · GROUP 11

# Medical Image Enhancement and Diagnostic Assistance System for Brain Tumor Detection

*Using MRI, OpenCV, classical machine learning and a fine-tuned CNN*

**SUBMITTED BY**

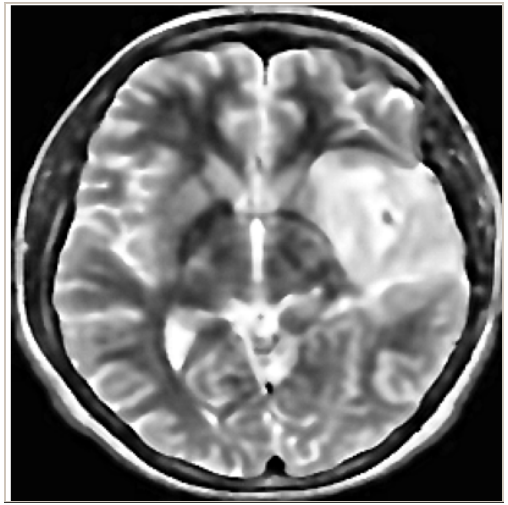
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**SESSION**

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*A test slice after CLAHE and unsharp masking, the system's enhancement stage.*

OVERVIEW

# Outline and Objectives


1. Problem statement
2. MRI as a signal
3. Image enhancement
4. System overview
5. Segmentation
6. Feature extraction
7. Classification
8. Detection results
9. Test-set audit
10. Tumor typing
11. Output for the clinician
12. Engineering challenges
13. Limitations, future scope
14. Conclusion


## Objectives

- Clean and enhance brain MRI slices with classical DSP filters, and measure the gain.
- Locate the suspicious region with a segmentation a person can follow step by step.
- Detect tumor vs no tumor, and identify the type: glioma, meningioma or pituitary.
- Show every intermediate stage, and test every claim on data the model never trained on.

# Problem Statement

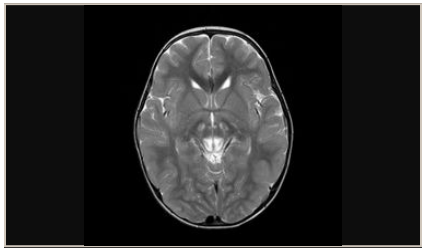
*The tumor is in the MRI, but noise and low contrast can hide it, and a label on its own gives a doctor nothing to check.*

## The image hides the lesion

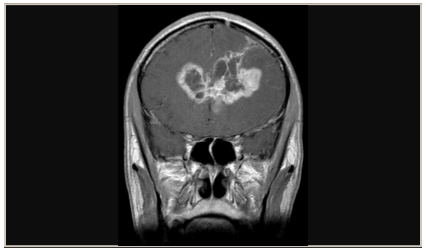
MRI slices carry noise, low contrast and unclear boundaries. A faint tumor next to the bright skull is easy to miss, and reading many scans tires even trained radiologists.

## A bare label cannot be verified

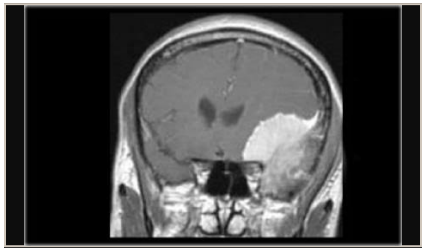
A classifier that only says “tumor, 97%” gives the radiologist nothing to inspect. A useful assistant has to show where it looked and what it measured.



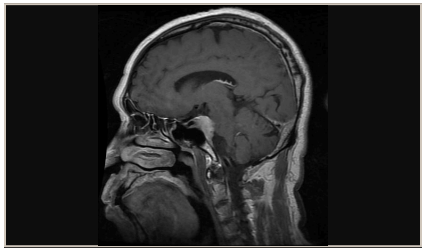
**No tumor** · 500 slices



**Glioma** · 926 · grows inside brain tissue



**Meningioma** · 937 · arises at the brain's rim



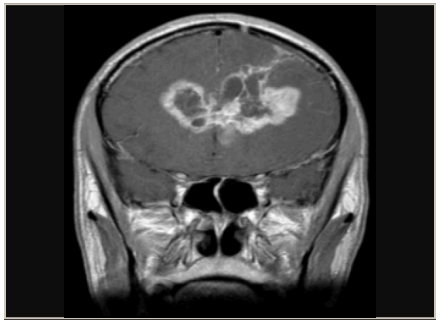
**Pituitary** · 901 · on the midline, at the skull base

# MRI as a 2-D Signal

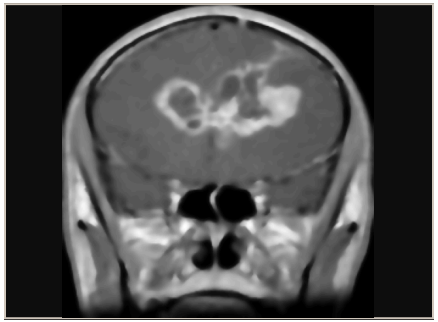
*Magnitude MRI noise is Rician, not Gaussian, and exported slices add impulse pixels, so each kind of noise gets its own filter.*

Rician magnitude: $m = \sqrt{((S + n_1)^2 + n_2^2)}, \quad n \sim \mathrm{N}(0, \sigma)$

Bilateral: $\mathrm{BF}[I]_p = (1/W_p)\ \Sigma_q\ G_{\sigma s}(\|p - q\|) \cdot G_{\sigma r}(|I_p - I_q|) \cdot I_q$



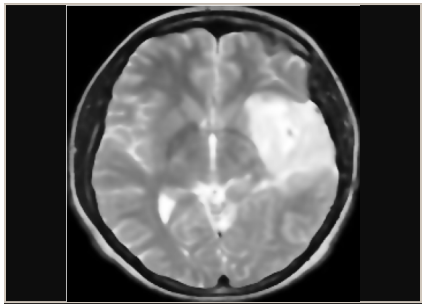
**As acquired**



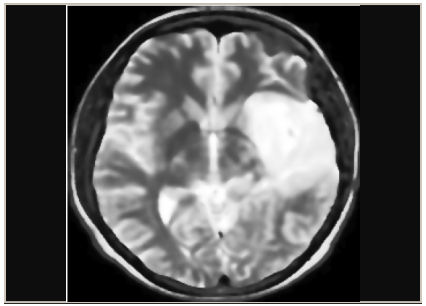
**After median + bilateral:** grain gone, edges kept

| # | Step | What it does |
|---|---|---|
| 1 | **Standardise** | Grayscale, crop blank borders, resize to 256×256 (area interpolation) |
| 2 | **Median 3×3** | Order statistic: an impulse never reaches the middle of the sorted window, so it vanishes without blurring edges |
| 3 | **Bilateral** | $d = 7$, $\sigma_{colour} = \sigma_{space} = 50$. The weight falls with distance *and* intensity difference, so it smooths inside tissue but stops at an edge |
| 4 | **Min–max** | Stretch to 0–255 so every scanner's slice starts from the same range |

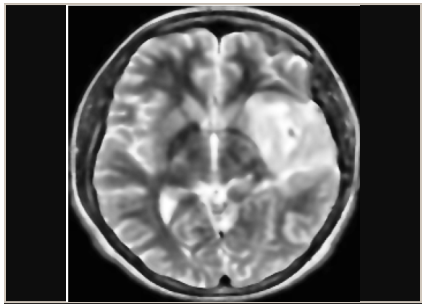
# Image Enhancement: CLAHE + Unsharp Masking



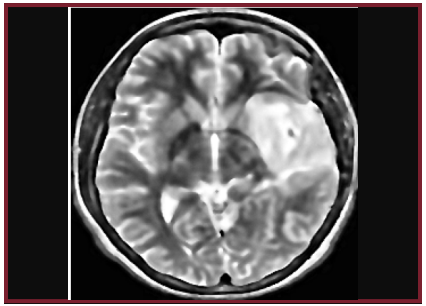
**(a) Preprocessed input**



**(b) Global hist. equalisation:** one mapping, dominated by skull and background



**(c) CLAHE:** 8×8 tiles, clip limit 2.0, noise held back by the clip



**(d) CLAHE + unsharp** (σ = 1.5, amount 1.2): the output we use

| Mean over the 394 test images | Preprocessed | Global HE | CLAHE only | CLAHE + unsharp |
|---|---|---|---|---|
| Contrast (std of intensities) | 57.9 | **75.6** | 65.3 | 70.4 |
| Entropy (bits) | 6.77 | 6.55 ↓ | **7.41** | 7.39 |
| Sharpness (variance of Laplacian) | 338 | 808 | 391 | **1428** |

Global HE gives the most raw contrast but loses information (entropy drops below the input). CLAHE + unsharp raises all three measures.

Measured with the project's enhancement_report() · enhance.py

# System Overview

*Seven stages take an uploaded slice to a verdict that a clinician can check.*

**S1 Data**
4-class MRI set, 3,264 slices; MRI phantom for tests

▶

**S2 Preprocess**
256×256, median + bilateral, normalise

▶

**S3 Enhance**
CLAHE + unsharp, gain measured

▶

**S4 Segment**
Skull strip, threshold, asymmetry, morphology

▶

**S5 Features**
171 numbers: intensity, texture, shape, location

▶

**S6 Classify**
SVM detects; SVM + ResNet-18 types

▶

**S7 Present**
Stage images, REST API, web app

## Tools

Python with OpenCV, scikit-image, scikit-learn and PyTorch; FastAPI service; Next.js web app.

## Data split

Bhuvaji et al. 4-class set with its own split: 2,870 training and 394 test slices.

## Design choice

Detection stays fully classical so its evidence can be inspected. The CNN is added only for typing, where the classical model fell short.

# Segmentation of the Suspicious Region

**1 — Skull strip**
Otsu head mask, fill holes, peel 22% of radius

**2 — Threshold**
Tissue above brain median: ≥ mean + 1.5σ

**3 — Asymmetry**
Mirror about the centroid; excess ≥ mean + 2σ

**4 — Morphology**
5×5 ellipse: open ×2, close ×3

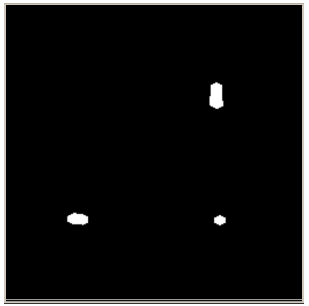
**5 — Contours**
Drop <120 px or aspect >5; keep the best

## Why not Otsu inside the brain?

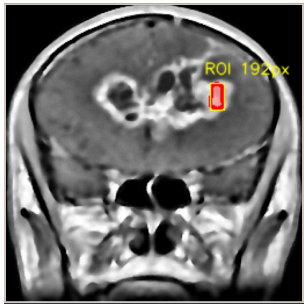
Otsu always splits a histogram in two, so on a healthy brain it reported grey vs white matter as a “lesion”. On 60 scans it flagged **100%** of healthy brains; the tissue-relative rule flagged **3%** and still caught every tumor.

## Why asymmetry?

A tumor breaks the brain's left–right symmetry even when it is not the brightest tissue. With both rules the pipeline localised **129 of 155** tumors on the 253-image Chakrabarty set.



**Mask** after morphology




**Region** outlined on the slice

# Feature Extraction

*Each scan becomes a vector of 171 numbers, grouped in seven families.*

| Family | Count | What it measures |
|---|---|---|
| Intensity + histogram | 33 | Mean, std, skew, kurtosis, entropy, percentiles; 16-bin histogram |
| GLCM texture | 38 | Contrast, homogeneity, energy, correlation at 1–3 px, 4 angles |
| LBP texture | 28 | Uniform local binary patterns, radius 3, 24 points |
| Shape of the region | 12 | Area, perimeter, circularity $4\pi A/P^2$, solidity, extent |
| Asymmetry | 10 | Statistics of the mirrored difference and its largest blob |
| Regional | 20 | Mean and std in a 3×3 grid over the brain |
| Location + plane | 30 | Distance from midline, depth, rim contact, scan-plane cues |
| **Total** | **171** | **Standardised inside the model pipeline** |

### Why location features?

Tumor type follows place: pituitary tumors sit on the midline, meningiomas at the rim, gliomas inside the tissue.

With texture alone, typing reached 72.1%. Adding position gave the classifier a cue a radiologist also uses.

# Classification Models

*Four classical models were compared on the same 171 features; a CNN joins only for tumor typing.*

| Model | Settings | CV ROC-AUC, detection | CV ROC-AUC, typing |
|---|---|---|---|
| Logistic regression | C = 1, balanced class weights (linear baseline) | 0.984 | 0.962 |
| **SVM, RBF kernel (selected)** | **C = 10, γ = scale, calibrated probabilities** | **0.993** | **0.983** |
| Random forest | 400 trees, balanced | 0.987 | 0.967 |
| MLP | Hidden layers 128 → 64, ReLU, $\alpha = 10^{-3}$ | 0.990 | 0.979 |
| ResNet-18 (typing only) | ImageNet weights, fine-tuned 14 epochs at 224×224, 4-view test-time averaging | – | – |

- Model selection by 5-fold stratified cross-validation on the **training split only**; scaling sits inside the pipeline so no fold leaks.
- Detection threshold **0.153**: the lowest cut that keeps out-of-fold specificity ≥ 80%, chosen for sensitivity.
- Typing blend: 0.95 weight on the CNN, chosen on a 431-image validation split.

# Results: Tumor Detection

*On the 394 held-out test slices, 271 of 289 tumors were found and no healthy scan was flagged.*

**95.4%**
accuracy

**93.8%**
sensitivity · 18 of 289 tumors missed

**100%**
specificity · 105 of 105 healthy cleared

**0.997**
ROC-AUC

| SVM-RBF, 171 features | Default cut | Tuned 0.153 |
|---|---|---|
| Missed tumors | 33 | **18** |
| False alarms | 0 | 0 |
| Accuracy | 91.6% | **95.4%** |
| Balanced accuracy | 94.3% | **96.9%** |

- In screening, a missed tumor costs far more than a second look, so the cut is tuned for **sensitivity**, on training data only.
- **Caution:** every healthy test slice has a near-copy in the training set, and all 18 missed tumors are slices without one. The next two slides explain this.

backend/outputs/metrics_binary_kaggle4.json

KEY FINDING · TEST-SET AUDIT

# 99% → 44%

Same typing model, same test folder. 259 of the 394 test slices have a near-identical twin in the training folder, and 88 are byte-for-byte copies. On those, the model scores 99.2%. On the 135 slices with no twin, it scores 43.7%.

Duplicate check: 64×64 normalised cross-correlation $\rho \geq 0.98$, following Vangala et al. (2026)

# Splitting the Test Set by “Seen Before”

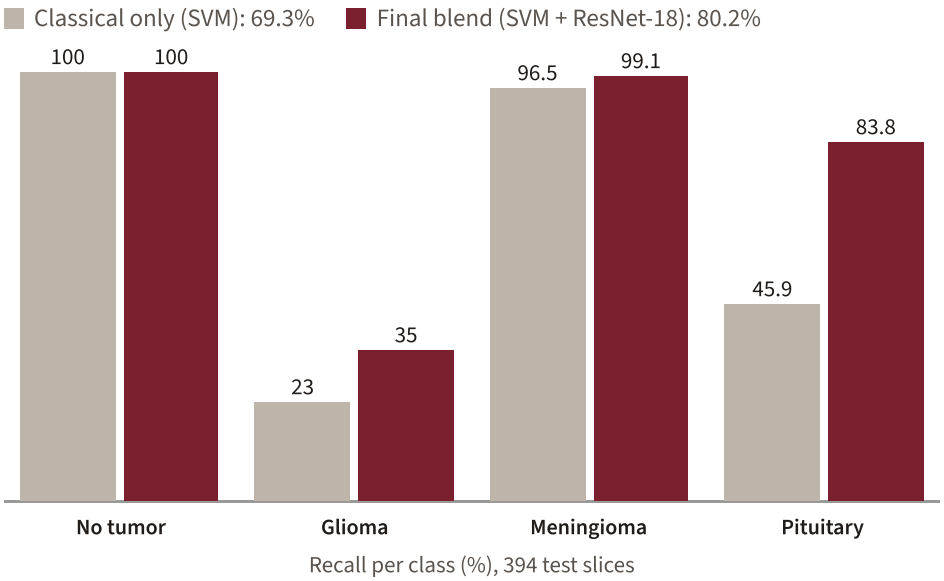
| Held-out test slices | Twin in training<br>259 slices | No twin<br>135 slices |
|---|---|---|
| Detection sensitivity | 100% (154/154) | **86.7%** (117/135) |
| Detection specificity | 100% (105/105) | not measurable |
| Typing accuracy | 99.2% | **43.7%** |
| Glioma recall | 90.0% (18/20) | **21.3%** (17/80) |
| Pituitary recall | 100% (25/25) | 75.5% (37/49) |
| Meningioma recall | 100% (109/109) | 83.3% (5/6) |

- **All 18 missed tumors** are slices with no twin in training.
- **Every healthy test slice has a twin**, so 100% specificity says nothing about a healthy brain the model has not seen.
- **Glioma is weak because it is new:** 80 of the 100 test gliomas have no twin, and every training glioma is 512×512 while test gliomas come in 64 sizes.
- Vangala et al. (arXiv:2610.00421) audited the same dataset and also found 259 near-twins.

Shipped models re-run on all 394 test slices; numbers match metrics_*_kaggle4.json

# Results: Tumor Typing

*The CNN lifts typing accuracy from 69.3% to 80.2%, but glioma remains the hard class.*




Recall per class (%), 394 test slices

| Actual ↓ / Predicted → | None | Gli | Men | Pit |
|---|---|---|---|---|
| No tumor | **105** | 0 | 0 | 0 |
| Glioma | **24** | **35** | **40** | 1 |
| Meningioma | 0 | 1 | **114** | 0 |
| Pituitary | 6 | 0 | 6 | **62** |

- Of 100 test gliomas, 40 were called meningioma and 24 “no tumor”.
- Macro F1 78.1%, one-vs-rest AUC 0.961.
- Validation split: 97.4% balanced accuracy; test folder: 79.5%.

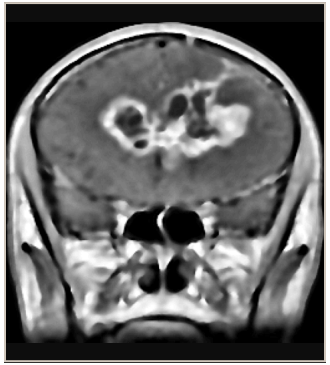
backend/outputs/metrics_multiclass_kaggle4.json

# Output for the Clinician

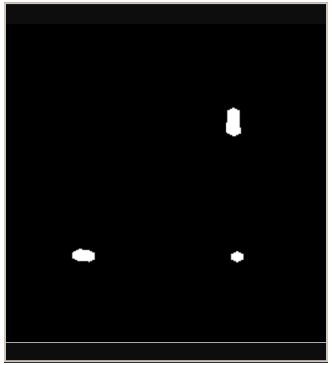
*Every stage is shown, so a doctor can see where the system looked and overrule it.*



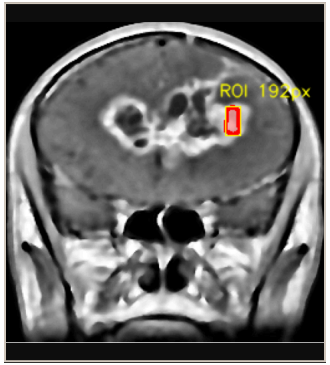
**Original**, as uploaded



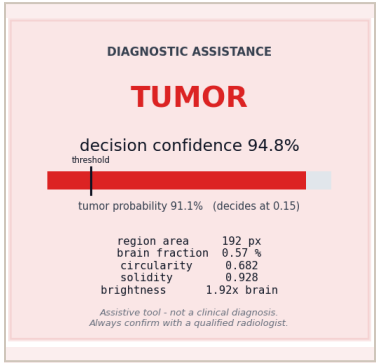
**Enhanced**, CLAHE + unsharp



**Mask**, threshold + morphology




**Region**, contour + box




**Verdict**: probability 91.1% vs threshold 0.15

**What the radiologist gets:** the verdict, the probability against the threshold, the region's area, circularity, solidity, brightness and location, and the enhancement gain. Uploads are never written to disk.

**Reading it critically:** here the outlined region (192 px) covers only one bright part of the lesion. The verdict comes from all 171 features, not the box, which is exactly why the box is shown.

# Engineering Challenges

*Four real bugs we found during development, each now covered by a regression test.*

| Problem | What went wrong | Fix |
|---|---|---|
| **Flood fill saw half the frame as a “hole”** | When the head touches the top and bottom borders, the background splits in two; the second half was filled in as brain. | Pad a 1-pixel black frame before the flood fill. |
| **Otsu flagged every healthy brain** | Otsu always finds two classes, so grey vs white matter became a “lesion”: 100% of healthy scans flagged. | Tissue-relative mean + 1.5σ rule: 3% flagged. |
| **Threshold tuning made results worse** | Maximising specificity picked 0.93 from over-confident training folds; test sensitivity fell from 91% to 71%. | Maximise sensitivity under a specificity floor. |
| **The ensemble peeked at the test set** | The first version kept whichever branch scored best on test, so the reported score was the best of three draws. | Choose the blend weight on a validation split. |

README, “Four real bugs”

# Limitations and Future Scope

## Limitations

- **Glioma typing is weak:** 35% recall; detection is strong, typing only partly solved.
- **A leaky public split:** 259 of 394 test slices have a near-twin in training; no patient IDs.
- **Specificity is untested** on healthy brains the model has not seen.
- **The CNN is a black box;** the audit trail comes from the classical branch only.
- **2-D slices only;** intensity rules can be fooled by calcification or artefacts.
- **Not a medical device:** clinical use needs multi-centre validation.

## Future scope

| Next step | Purpose |
|---|---|
| **De-duplicate and re-split** | Honest test set; add a second dataset (Figshare, Cheng et al.) |
| Patient-level splits | No patient on both sides of the split |
| Grad-CAM on the CNN | Make the typing decision checkable too |
| U-Net with Dice scoring | Pixel-level masks, measured overlap |
| 3-D tumor volume | Combine slices of a full MRI stack |
| PDF reports | Stages, measurements and verdict in one document |

README, “Limitations” and “Future scope”

# Conclusion

We built a working pipeline from an uploaded MRI slice to an annotated verdict, with every stage visible through a REST API and a web app. Three lessons stand out:

## 01

### Enhance locally

CLAHE + unsharp masking raised entropy from 6.77 to 7.39 bits and sharpness more than four times.

## 02

### Tune away from the test set

A sensitivity-first threshold cut missed tumors from 33 to 18 with no false alarms.

## 03

### Check the test set for twins

On the 135 slices never seen before, typing falls from 99% to 44%: the honest estimate.

| Headline numbers | Detection accuracy | Sensitivity | Specificity | Typing accuracy |
|---|---|---|---|---|
| Full public test split (394 slices) | 95.4% | 93.8% | 100% | 80.2% |
| **Slices never seen before (135)** | **–** | **86.7%** | **not measurable** | **43.7%** |

The second row is the estimate we would trust for new patients.

# References


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# Thank You

*Questions and feedback are welcome*

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